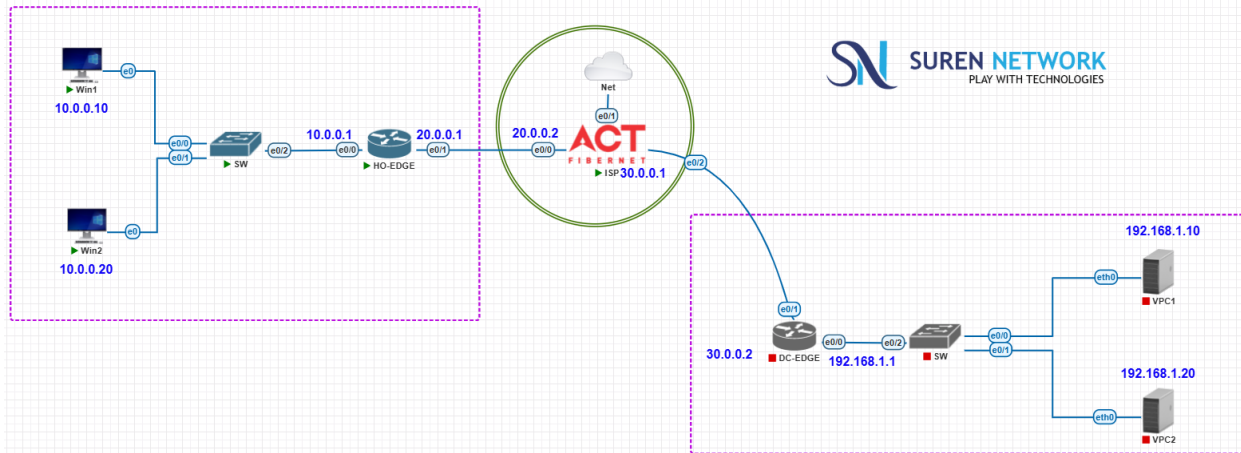


NAT



Step 1 : Assign ip address for HO router

```
HO-EDGE
HO-EDGE(config)#do sh ip int bri
Interface          IP-Address      OK? Method Status  Prot
Ethernet0/0        10.0.0.1        YES manual up      up
Ethernet0/1        20.0.0.1        YES manual up      up
Ethernet0/2        unassigned      YES TFTP  down    down
Ethernet0/3        unassigned      YES TFTP  down    down
```

Step 2 : Add default route in head office router

```
HO-EDGE(config)#ip rou
HO-EDGE(config)#ip route 0.0.0.0 0.0.0.0 20.0.0.2
HO-EDGE(config)#
```

Step 3 : Enable dhcp in HO router for lan pc

```
HO-EDGE
HO-EDGE(config)#ip dhcp pool LAN
HO-EDGE(dhcp-config)#network 10.0.0.0 255.0.0.0
HO-EDGE(dhcp-config)#default-route 10.0.0.1
HO-EDGE(dhcp-config)#dns-server 8.8.8.8
HO-EDGE(dhcp-config)#exit
HO-EDGE(config)#
```

```
QEMU (Win2)
C:\WINDOWS\system32\cmd.exe
Ethernet adapter Local Area Connection:

    Connection-specific DNS Suffix  . : 
    IP Address . . . . . : 10.0.0.3
    Subnet Mask . . . . . : 255.0.0.0
    Default Gateway . . . . . : 10.0.0.1
```

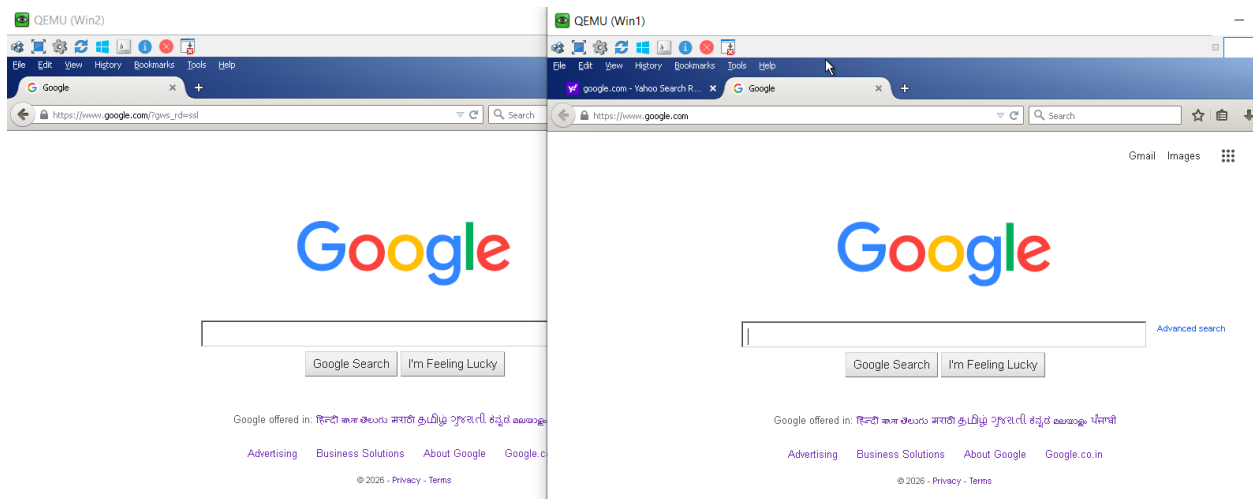
```
QEMU (Win1)
C:\WINDOWS\system32\cmd.exe
Ethernet adapter Local Area Connection:

    Connection-specific DNS Suffix  . : 
    IP Address . . . . . : 10.0.0.2
    Subnet Mask . . . . . : 255.0.0.0
    Default Gateway . . . . . : 10.0.0.1
```

Step 4 : Create dynamic overload nat in head office router

```
HO-EDGE
HO-EDGE(config)#interface e0/0
HO-EDGE(config-if)#ip nat inside
HO-EDGE(config-if)#ex
HO-EDGE(config)#
*Sep 12 11:39:33.143: %LINEPROTO-5-UPDOWN: Line protocol on Interface M
ged state to upex
HO-EDGE(config)#interface e0/1
HO-EDGE(config-if)#ip nat outside
HO-EDGE(config-if)#exit
HO-EDGE(config)#
HO-EDGE(config)#access-list 10 permit 10.0.0.0 0.255.255.255
HO-EDGE(config)#ip nat inside source list 10 interface e0/1 overload
```

Step 5 : After getting ip from dhcp access internet



Step 6 : Create static nat in datacenter router

```
DC-EDGE
Router(config)#int e0/0
Router(config-if)#ip nat inside
Router(config-if)#
*Sep 12 12:24:36.274: %LINEPROTO-5-UPDOWN: Line protocol on Interface
ged state to up
Router(config-if)#exit
Router(config)#int e0/1
Router(config-if)#ip nat outside
Router(config-if)#ex
Router(config)#
Router(config)#ip nat inside source static 192.168.1.10 30.0.0.10
Router(config)#ip nat inside source static 192.168.1.20 30.0.0.11
Router(config)#
Router(config)#do wr
Building configuration...
[OK]
```

Step 7 : Access datacenter server from head office pc

QEMU (Win1)	QEMU (Win2)
<pre>C:\WINDOWS\system32\cmd.exe C:\Documents and Settings\user>ping 30.0.0.10 Pinging 30.0.0.10 with 32 bytes of data: Reply from 30.0.0.10: bytes=32 time=5ms TTL=61 Reply from 30.0.0.10: bytes=32 time=3ms TTL=61 Reply from 30.0.0.10: bytes=32 time=2ms TTL=61 Reply from 30.0.0.10: bytes=32 time=2ms TTL=61 Ping statistics for 30.0.0.10: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 2ms, Maximum = 5ms, Average = 3ms C:\Documents and Settings\user> C:\Documents and Settings\user> C:\Documents and Settings\user> C:\Documents and Settings\user> C:\Documents and Settings\user> C:\Documents and Settings\user></pre>	<pre>C:\WINDOWS\system32\cmd.exe Microsoft Windows XP [Version 5.1.2600] (C) Copyright 1985-2001 Microsoft Corp. C:\Documents and Settings\user>color f0 C:\Documents and Settings\user>ping 30.0.0.2 Pinging 30.0.0.2 with 32 bytes of data: Reply from 30.0.0.2: bytes=32 time=4ms TTL=253 Reply from 30.0.0.2: bytes=32 time=2ms TTL=253 Reply from 30.0.0.2: bytes=32 time=3ms TTL=253 Reply from 30.0.0.2: bytes=32 time=2ms TTL=253 Ping statistics for 30.0.0.2: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 2ms, Maximum = 4ms, Average = 2ms C:\Documents and Settings\user></pre>